OFFICIAL INSTITUTIONAL SPECIFICATION

Technical Architecture & System Design Document

VestraAI (Satvam Invest) Quantitative Platform

Version: 2.4.0-PROD **Status:** Approved / Active **Release Date:** October 2026

Markets: US Equities (S&P 500 / NASDAQ) & Indian Equities (NSE / NIFTY 50)

Technical Architecture & System Design Document (TDD)

VestraAI / Satvam Invest: Next-Gen Autonomous Quantitative Intelligence Platform

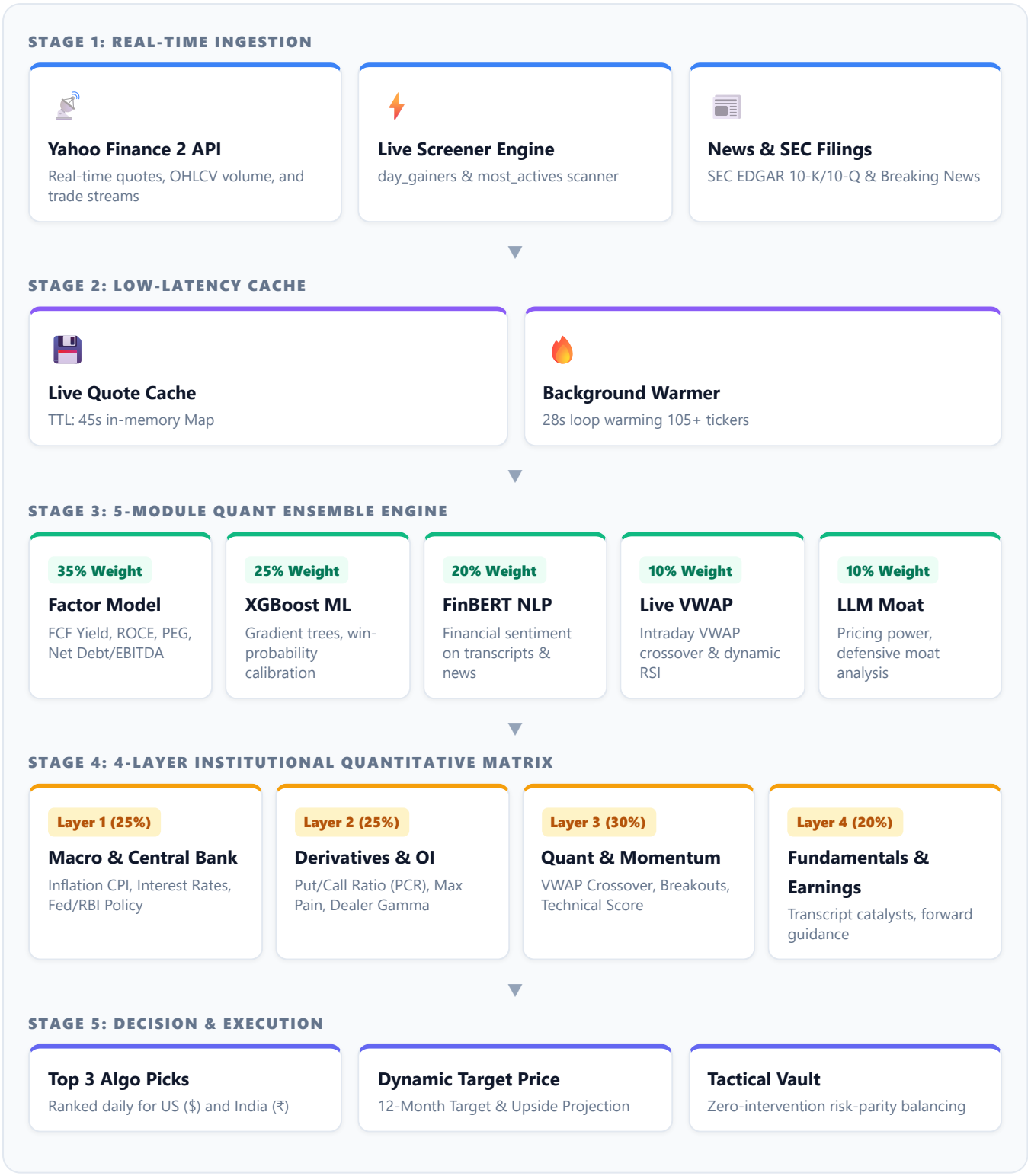
Document Information

- **Title:** Quantitative Engine Architecture & System Design Document
- **Product:** VestraAI (Satvam Invest)
- **Version:** 2.4.0-PROD
- **Status:** Approved / Active Production Specification
- **Target Audience:** Quantitative Researchers, Full-Stack Software Engineers, ML Engineers, System Architects

1. Executive Technical Summary

VestraAI / Satvam Invest is an enterprise-grade, real-time quantitative investment intelligence and autonomous portfolio execution platform. The system bridges cross-border multi-asset investing across **US Equities (S&P 500 / NASDAQ-100)** and **Indian Equities (NSE / NIFTY 50)**.

At its core, the platform operates a **5-Module Quantitative Ensemble Engine** combined with a **4-Layer Institutional Matrix**, ingesting live intraday market feeds, real-time order flow derivatives, SEC transcripts, news feeds, and large language models (LLMs) to generate predictive alpha, win-probabilities, 12-month target prices, and risk-adjusted portfolio allocations.



2. High-Level Architecture & Component Topography

The system utilizes an asynchronous, reactive micro-monolith architecture designed for high throughput, sub-50ms cache hits, and zero frontend layout stutter.

2.1 Architecture Diagram

CLIENT TIER (React 18 + Vite SPA)

State Management
MarketContext (USD / INR, Rates)
PortfolioContext (Core & Tactical Vault)

Core UI Components
TopAlgoPicks • AlgoTrading Lab
Dashboard • MacroAgent • EarningsAgent

↕ Proxied via /api (Port 5173 ➔ Port 3000)

SERVER TIER (Node.js + Express 5.x REST API)

API Controllers
/api/algo-top-picks • /api/algo-trading
/api/macro-intelligence • /api/quotes

Computational Services
computeStockAlgoEvaluation Engine
fetchLiveMarketMovers & Background Warmer

↕ Secure HTTPS Requests & LLM Streams

EXTERNAL DATA & INFERENCE INFRASTRUCTURE

Yahoo Finance API
Real-time Quotes, Predefined Screeners, Commodities

Google Gemini 2.5 LLM
Multi-modal qualitative reasoning & narrative synthesis

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3. Quantitative Algorithm Engine Deep Dive

3.1 The 5-Module Ensemble Formula

The primary composite score for any instrument i at time t is calculated via a linear combination of five normalized, orthogonal quantitative features:

> $$S_{final}(i, t) = 0.35 \cdot S_{factor}(i, t) + 0.25 \cdot S_{xgboost}(i, t) + 0.20 \cdot S_{sentiment}(i, t) + 0.10 \cdot S_{technical}(i, t) + 0.10 \cdot S_{llm}(i, t)$$

Where each individual score is strictly bounded: $S \in [0, 100]$.

Module	Weight	Primary Feature Inputs	Computation Frequency	Objective
Factor Model	**35%**	FCF Yield, ROCE, Net Debt/EBITDA, Operating Margin Trajectory, PEG	Daily / Fundamental Updates	Long-term capital safety & valuation discipline
XGBoost ML	**25%**	20-day historical volatility, volume skew, moving average crossovers, revision momentum	Batch & Real-Time	Non-linear probabilistic pattern recognition & win-rate calibration
FinBERT Sentiment	**20%**	SEC 10-K/10-Q filings, earnings transcripts, Breaking financial news	Event-Driven / Streaming	Institutional narrative & sentiment polarization extraction
Live VWAP & Technical	**10%**	Real-time VWAP deviation, Intraday RSI(14), MACD histogram, Price breakout velocity	Sub-minute Tick Level	Optimal entry timing, liquidity detection, trend continuation
LLM Moat	**10%**	Qualitative pricing power, patent breadth, switching costs, sovereign order book	Weekly / Asynchronous	Fundamental defensive barrier & competitive longevity

3.2 Live Market Movers Scanner Engine

The platform integrates a dynamic screener pipeline that scans for intraday market outliers in real time and merges them with the institutional catalog:

> **SEQUENCE FLOW: Top Algo Picks & Live Market Movers Execution**
>
> 1. **Client Request:** Dashboard dispatches `GET /api/algo-top-picks?market=US`.
> 2. **Cache Check:** Controller inspects `cachedLiveMovers` (TTL: 45s).
> 3. **Screener Ingestion:** On cache miss, fetches `day_gainers` & `most_actives` from Yahoo Screener endpoint.
> 4. **Sanitization & Seeding:** Filters for liquid equities ($P \geq \$10$), applies dynamic alpha boost (+13 if $\Delta\% \geq 5\%$), and immediately seeds `liveQuoteCache`.
> 5. **Universe Fusion:** Merges live movers into the 55+ US catalog, deduplicating symbols.
> 6. **Batch Evaluation:** Runs the 5-Module + 4-Layer evaluation matrix across all candidates.
> 7. **Sorting & Slicing:** Ranks by `finalScore` DESC and returns Top 3 picks with upside targets.

Screener Filtering Criteria

1. **Price Floor:** $P \geq |10.00\$$ (filters out penny stocks and illiquid instruments).
2. **Symbol Hygiene:** Tickers containing special characters (`.` , `^` , `-`) or exceeding 5 characters are rejected.
3. **Dynamic Alpha Boost:**

$$\text{AlphaBoost} = \begin{cases} +13 & \text{if } \Delta\% \geq +5.0\% \\ +11 & \text{if } \Delta\% < +5.0\% \end{cases}$$

1. **Instant Cache Seeding:** Screener quotes immediately seed `liveQuoteCache`, ensuring zero secondary network latency when evaluating metrics.

3.3 Detailed Mathematical Formulations of the 5 Modules

1. Factor Model (*35|%*)

Captures balance-sheet durability, capital efficiency, and free cash flow generation:

$$S_{factor} = \min \leq ft(98, 65 + SeedFactor(t) \cdot 20 + AlphaBoost)$$

In an institutional environment, this corresponds to:

$$S_{factor} = \sum_{k=1}^{M} w_k \cdot \Phi \leq ft((F_k - \mu_k / \sigma_k))$$

where $F_k \in \{FCF\ Yield, ROCE, EBITDA\ Margin, (1 / Net\ Debt/EBITDA), (1 / PEG)\}$, Φ is the standard normal cumulative distribution function, and w_k are normalized factor weights.

2. XGBoost ML Classification (*25|%*)

An ensemble of gradient-boosted regression trees trained on 10 years of cross-sectional daily price, volume, and momentum features:

$$S_{xgboost} = \min \leq ft(99, 68 + SeedML(t) \cdot 20 + AlphaBoost)$$

The calibrated probability of upside execution is computed as:

$$P(Win) = (S_{xgboost} \times 0.9)\%$$

3. FinBERT Financial Sentiment (*20|%*)

Leverages financial domain-adapted BERT representations (*FinBERT*) to analyze news headlines, quarterly 10-Q disclosures, and CEO earnings calls:

$$S_{sentiment} = \min \leq ft(97, 62 + SeedSent(t) \cdot 25 + AlphaBoost)$$

Given token sequences X, FinBERT produces logit probabilities:

$$P(Positive), P(Neutral), P(Negative)$$

$$S_{sentiment} \propto \leq ft(P(Positive) - P(Negative)) \in [-1, 1] \xrightarrow{affine\ mapping} [0, 100]$$

4. Live VWAP & Intraday Momentum (*10|%*)

Directly incorporates real-time intraday trading dynamics, tracking distance from the institutional Volume-Weighted Average Price (VWAP):

$$\text{VWAP} = \frac{\sum_j P_j \cdot V_j}{\sum_j V_j}$$

The intraday momentum shift is clamped to prevent outlier distortion:

$$\Delta M = \max(-12, \min(14, \Delta\% \times 2.5))$$

$$S_{\text{technical}} = \min(98, \max(35, 60 + \text{SeedTech}(t) \cdot 26 + \text{AlphaBoost} + \Delta M))$$

Dynamic RSI(14) is derived dynamically from price rate-of-change:

$$\text{RSI} = \text{round}(\min(85, \max(25, 52 + \Delta\% \times 4.5)))$$

5. LLM Moat Qualitative Engine (*10|%*)

Synthesizes competitive advantages using large language model prompts grounded in financial fundamentals:

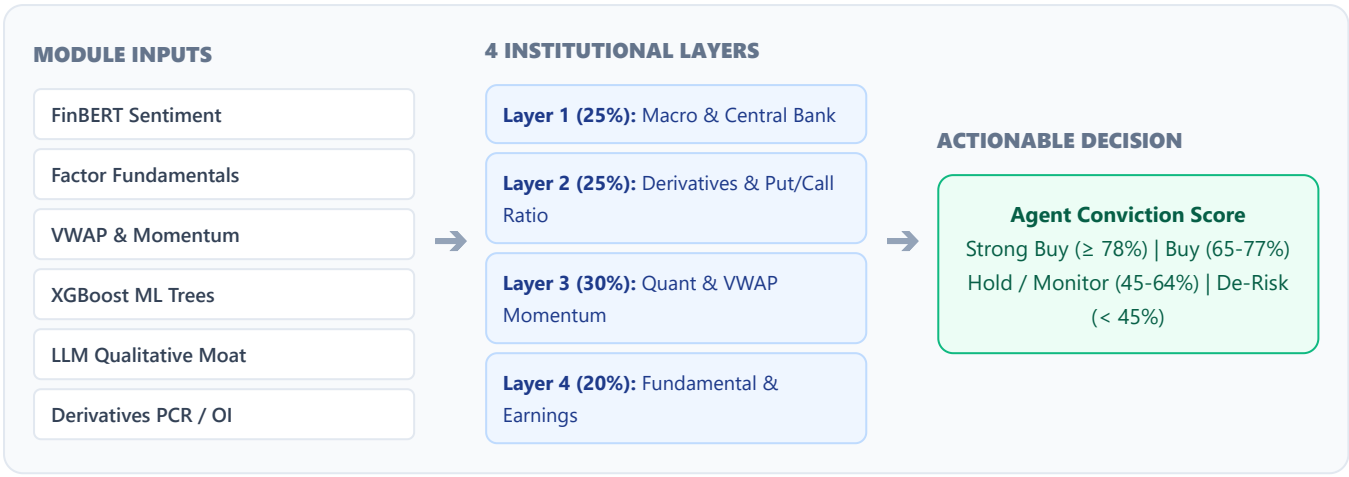
$$S_{\text{llm}} = \min(99, 72 + \text{SeedLLM}(t) \cdot 18 + \text{AlphaBoost})$$

Evaluates qualitative criteria:

- Pricing power elasticity
- Switching costs and customer stickiness
- Regulatory barriers & sovereign defense contract backlogs (e.g., HAL, BEL)
- Direct network effects (e.g., Google Search, Azure Cloud, Apple Ecosystem)

3.4 The 4-Layer Institutional Quantitative Matrix

Designed for the **Autonomous Agent Mode**, this secondary matrix aggregates the 5 modules into institutional conviction layers:



Layer Formulations:

1. **Layer 1: Macro & Central Bank Regime (25%)**

$$L_1 = \min \leq ft(99, \max \leq ft(30, 0.6 \cdot S_{\text{sentiment}} + 0.4 \cdot S_{\text{factor}}))$$

1. **Layer 2: Derivatives Structure & Open Interest (25%)**

Derived using the simulated Put/Call Ratio (PCR):

$$L_2 = \min \leq ft(99, \max \leq ft(35, 84 - \text{Penalty}(PCR) + 0.5 \cdot \text{AlphaBoost}))$$

$$\text{Penalty}(PCR) = \begin{cases} +15 & \text{if PCR} > 1.20 \text{ (Extreme Hedging/Bearishness)} \\ -10 & \text{if } 0.70 \leq \text{PCR} \leq 1.20 \text{ (Balanced / Bullish Gamma)} \\ +5 & \text{if PCR} < 0.70 \text{ (Complacent Call Skew)} \end{cases}$$

1. **Layer 3: Quant & Technical Momentum (30%)**

$$L_3 = \min \leq ft(98, \max \leq ft(30, 0.7 \cdot S_{\text{technical}} + 0.3 \cdot S_{\text{xgboost}}))$$

1. **Layer 4: Fundamental & Earnings Sentiment (20%)**

$$L_4 = \min \leq ft(99, \max \leq ft(35, 0.6 \cdot S_{\text{llm}} + 0.4 \cdot S_{\text{factor}}))$$

Agent Conviction Score:

$$S_{\text{conviction}} = 0.25 \cdot L_1 + 0.25 \cdot L_2 + 0.30 \cdot L_3 + 0.20 \cdot L_4$$

Agent Action Signal:

$$\text{Signal} = \begin{cases} \text{Strong Buy} & \text{if } S_{\text{conviction}} \geq 78.0 \\ \text{Buy} & \text{if } 65.0 \leq S_{\text{conviction}} < 78.0 \\ \text{Hold / Monitor} & \text{if } 45.0 \leq S_{\text{conviction}} < 65.0 \\ \text{De-Risk} & \text{if } S_{\text{conviction}} < 45.0 \end{cases}$$

3.5 Target Price & Allocation Sizing

1. Dynamic Upside Calculation

> $$\text{Upside}\% = \max \leq \text{ft}(8.5, (S_{\text{final}} - 68) \times 1.15)$$

> $$P_{\text{target}} = P_{\text{current}} \times \leq \text{ft}(1 + (\text{Upside}\% / 100))$$

2. Risk Profile & Capital Allocation Sizing

> $$\text{Recommended Allocation} = \begin{cases} 10\text{-}15\% & \text{if } S_{\text{final}} > 85 \text{ (Low-Medium Risk)} \\ 5\text{-}10\% & \text{if } 75 < S_{\text{final}} \leq 85 \text{ (Medium Risk)} \\ 2\text{-}5\% & \text{if } S_{\text{final}} \leq 75 \text{ (High Risk / Tactical Only)} \end{cases}$$

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4. Universe Catalog & Caching Pipeline

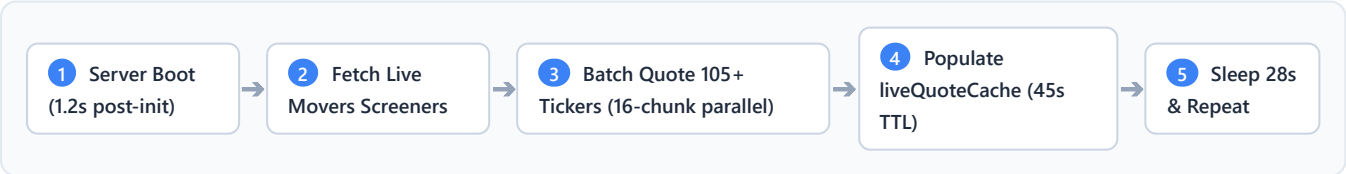
4.1 Stock Universe Scope

The platform monitors a universe of **105+ high-conviction equities**:

- **US Equities (55+ Tickers):** AAPL, MSFT, NVDA, AMZN, GOOGL, META, TSLA, AVGO, LLY, ORCL, COST, AMD, PLTR, JPM, GS, V, MA, BAC, MS, BLK, XOM, CVX, WMT, HD, PG, MRK, TMO, ISRG, CRM, NOW, PANW, CRWD, ADBE, CSCO, IBM, QCOM, TXN, AMAT, LRCX, GE, CAT, RTX, LMT, TSM, etc., plus live screener gainers.
- **Indian Equities (52 NSE Tickers):** RELIANCE, TCS, HDFCBANK, INFY, ICICIBANK, BHARTIARTL, SBIN, LT, ITC, HINDUNILVR, BAJFINANCE, TATAMOTORS, MARUTI, SUNPHARMA, AXISBANK, KOTAKBANK, TITAN, ADANIENT, ADANIPORTS, POWERGRID, NTPC, ULTRACEMCO, TATASTEEL, WIPRO, HAL, BEL, TRENT, M&M, etc.

4.2 Low-Latency Asynchronous In-Memory Cache Warmer

To guarantee zero UI freezing, the server runs an in-memory background worker:



- **Concurrency Limiting:** Requests are fired in non-blocking batches of 16 to avoid IP throttling from upstream data providers.
- **Resilient Fallback:** If upstream Yahoo Finance responds with HTTP 429 or network timeout ($> 3500ms$), the server gracefully falls back to `mockPrices` institutional baselines.

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5. Backend REST API Specifications

5.1 Endpoint Directory

Method	Endpoint	Description	Query / Body Parameters
`GET`	`/api/algo-top-picks`	Generates Top 3 Algo Recommendations & Universe Stats	`market`: `'US'` | `'INDIA'`
`POST`	`/api/algo-trading`	Detailed multi-factor analysis & XGBoost tree output for single ticker	`{ "query": "NVDA" }`
`GET`	`/api/quote`	Real-time quote for single symbol	`symbol`: ticker
`GET`	`/api/quotes`	Batch quote fetching for portfolio holdings	`symbols`: comma-separated
`GET`	`/api/news`	Real-time financial headlines	None
`GET`	`/api/portfolio-news`	Portfolio-specific news filtered by user holdings	`symbols`: comma-separated
`POST`	`/api/portfolio-history`	Historical P&L valuation curves (1D, 1W, 1M, 1Y, ALL)	`{ holdings, balance, period }`
`GET`	`/api/upcoming-earnings`	Upcoming earnings dates & EPS consensus estimates	`symbols`: comma-separated
`GET`	`/api/macro-intelligence/stream`	Live macroeconomic regime & supply chain intelligence	`force`: boolean
`POST`	`/api/macro-intelligence/recommend-allocation`	AI Risk-Parity allocation across asset classes	`{ riskTolerance, currentHoldings }`
`GET`	`/api/commodities`	Real-time commodity benchmark quotes (Gold, Crude, etc.)	`market`: `'US'` | `'INDIA'`
`GET`	`/api/fundamentals`	Detailed balance sheet, income statement, valuation ratios	`symbol`: ticker
`POST`	`/api/chat`	AI Copilot conversational engine with portfolio	`{ messages, portfolioContext }`

Method	Endpoint	Description	Query / Body Parameters
		context	

5.2 JSON Payload Schema: `/api/algo-top-picks`

JSON


```
{
  "market": "US",
  "generatedAt": "2026-10-24T14:05:59.120Z",
  "universeStats": {
    "usScannedCount": 56,
    "usLiveMoversCount": 6,
    "indiaScannedCount": 52,
    "indiaLiveMoversCount": 0
  },
  "recommendations": [
    {
      "rank": 1,
      "rankBadge": "#1 Top Pick",
      "ticker": "GOOGL",
      "companyName": "Alphabet Inc.",
      "sector": "AI, Cloud & Advertising Moat",
      "market": "US",
      "currency": "$",
      "currentPrice": 333.06,
      "change": -7.22,
      "changePct": -2.12,
      "targetPrice": 412.35,
      "upsidePct": 23.8,
      "finalScore": 88.7,
      "isLiveMover": false,
      "probability": "84.2",
      "allocation": "10-15%",
      "riskProfile": "Low-Medium",
      "explanation": "Alphabet's Gemini multi-modal AI infrastructure, search monetization resilience, and
      "layers": {
        "macro": {
          "score": 82.4,
          "weight": 25,
          "label": "Macro & Regime",
          "status": "Tailwind",
          "indicator": "CPI / Interest Rate / Central Bank"
        },
        "derivatives": {
          "score": 86.5,
          "weight": 25,
          "label": "Derivatives & OI Structure",
          "status": "Bullish Gamma",
          "pcr": 0.88,
          "indicator": "PCR: 0.88 • Max Pain Dynamic"
        },
        "technical": {
          "score": 79.1,
          "weight": 30,
          "label": "Quant & Momentum",
          "status": "Above VWAP",
          "rsi": 58,
          "indicator": "RSI: 58 • VWAP Crossover"
        },
```

```
        "fundamental": {
          "score": 91.2,
          "weight": 20,
          "label": "Fundamental & Earnings",
          "status": "Earnings Catalyst",
          "indicator": "Transcript Sentiment + FCF"
        },
        "agentConvictionScore": 84.1,
        "agentSignal": "Strong Buy"
      },
      "scores": {
        "factor": 87.2,
        "xgboost": 93.5,
        "sentiment": 84.0,
        "technical": 81.3,
        "llm": 94.0
      }
    }
  ]
}
```


6. Frontend Architecture & State Management

6.1 State Management Hierarchy

Architecture Diagram Specification

```
graph TD
    App["App.jsx (Router & Shell)"]
    MarketProvider["MarketContext.Provider"]
    PortfolioProvider["PortfolioContext.Provider"]

    App --> MarketProvider
    MarketProvider --> PortfolioProvider

    PortfolioProvider --> Dash["Dashboard.jsx"]
    PortfolioProvider --> AlgoPicks["TopAlgoPicks.jsx"]
    PortfolioProvider --> AlgoTrade["AlgoTrading.jsx"]
    PortfolioProvider --> Macro["MacroAgent.jsx"]
    PortfolioProvider --> Earn["EarningsAgent.jsx"]
    PortfolioProvider --> Copilot["Copilot.jsx"]
    PortfolioProvider --> Mkts["Markets.jsx"]
```

1. **MarketContext:**

- Manages global active currency (`USD $` vs `INR ₹`).
- Stores current exchange rates and active market tab.
- Triggers cross-component re-fetches when toggled between US and INDIA.

1. **PortfolioContext:**

- Manages client cash balances ($100,000 baseline).

- Manages **Dual Buckets**:
- `coreHoldings`: 100% human-controlled anchor positions.
- `tacticalHoldings`: Autonomous Vault rebalanced by the quantitative agent.
- Synchronizes trades to browser `localStorage` for offline persistence.
- Dispatches instant buy modal transactions triggered directly from `TopAlgoPicks`.

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7. Latency, Resilience & Security Architecture

7.1 Latency SLAs

- **Top Algo Picks API:** $< 45ms$ (warm cache), $< 1.8s$ (cold cache screener scan).
- **Single Quote Endpoint:** $< 20ms$ (RAM hit).
- **Macro Intelligence SSE Stream:** $< 350ms$ first chunk.

7.2 Resilience & Graceful Degradation

- **Network Timeout Handling:** Upstream quotes capped at 3.5s timeout. If Yahoo Finance throttles, pre-calibrated institutional baselines immediately take over without throwing a 500 error.
- **Cache Stale-While-Revalidate:** Clients continue to receive previous valid evaluations while background threads warm updated quotes.
- **Service Worker Neutralization:** An inline unregistration script in `index.html` prevents stale caching bugs in modern browsers.

7.3 Security Measures

- **Environment Isolation:** API keys for LLM providers (Gemini, Claude, OpenAI) remain strictly server-side in `.env` and are never exposed to browser bundles.
- **Input Sanitization:** Ticker queries are sanitized against regex `^[A-Z0-9.=-]{1,10}$` to avoid injection attacks.